

① Perturbation Theory (General Approach).

$$H = H_0 + V \quad (V = \alpha H_1)$$

$$H_0 |n^0\rangle = E_n^0 |n^0\rangle$$

$$H |n\rangle = E_n |n\rangle$$

We focus on a degenerate subspace with projection operator $\hat{P}_D$ and ask how these states change under the action of the perturbation. Thus

$$\hat{P}_D |n\rangle = |\tilde{n}^0\rangle \quad (\text{in the notation of the class})$$

$$|n\rangle = |\tilde{n}^0\rangle + \alpha |n^1\rangle + \alpha^2 |n^2\rangle + \dots$$

$$E_n = E_n^0 + \alpha E_n^1 + \alpha^2 E_n^2 + \dots$$

we call $\mathcal{E} = E_n^0$ and

$$\Delta_n = \alpha E_n^1 + \alpha^2 E_n^2 + \dots$$

Thus we have

$$H |n\rangle = (\mathcal{E} + \Delta_n) |n\rangle$$

$$\Rightarrow (H_0 + V) |n\rangle = (\mathcal{E} + \Delta_n) |n\rangle$$

But note that $|n\rangle = \hat{P}_D |n\rangle + \hat{\Phi}_D |n\rangle$

(2)

Hence :

$$\begin{aligned}
H_0 \hat{P}_D |n\rangle + H_0 \hat{\Phi}_D |n\rangle + V P_D |n\rangle + V \Phi_D |n\rangle \\
= \epsilon P_D |n\rangle + \epsilon \Phi_D |n\rangle + \Delta_n P_D |n\rangle \\
+ \Delta_n \Phi_D |n\rangle
\end{aligned}$$

— (1)

(a) Applying P_D on both sides of eq. (1) we get-

$$\begin{aligned}
P_D H_0 P_D |n\rangle + P_D V P_D |n\rangle + P_D V \Phi_D |n\rangle \\
= \epsilon P_D |n\rangle + \Delta_n P_D |n\rangle .
\end{aligned}$$

Note: we have used $P_D H_0 \Phi_D = 0$ $P_D \Phi_D = 0$

Now $P_D H_0 P_D = \epsilon P_D$, so we get

$$P_D V P_D |n\rangle + P_D V \Phi_D |n\rangle = \Delta_n P_D |n\rangle .$$

(b) Applying Φ_D on both sides of eq. (1) we get-

$$\Phi_D H_0 \Phi_D |n\rangle + \Phi_D (V - \Delta_n) |n\rangle = \epsilon \Phi_D |n\rangle$$

$$\Rightarrow \Phi_D (\epsilon - H_0) \Phi_D |n\rangle = \Phi_D (V - \Delta_n) |n\rangle .$$

Now $\Phi_D (\epsilon - H_0) \Phi_D$ is invertable (no. zero eigenvalues!)

$$\text{So } \Phi_D |n\rangle = \frac{\Phi_D}{\epsilon - H_0} (V - \Delta_n) |n\rangle .$$

(3)

(c) From part (b) we see

$$\phi_D |n\rangle = \frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) P_D |n\rangle + \frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) \phi_D |n\rangle$$

Substituting $\phi_D |n\rangle$ on the R.H.S repeatedly one obtains

$$\phi_D |n\rangle = \left[\frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) P_D |n\rangle + \frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) \frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) P_D |n\rangle + \dots \right]$$

$$= \left[\frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) + \frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) \frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) + \dots \right] P_D |n\rangle.$$

(d) Substituting the iterative soln. of (c) into (a) we get

$$P_D V \left\{ 1 + \frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) + \frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) \frac{\phi_D}{\varepsilon - H_0} (V - \Delta_n) + \dots \right\} P_D |n\rangle = \Delta_n P_D |n\rangle.$$

which is an eigenvalue eq. which when solved gives $P_D |n\rangle$ and Δ_n .

(c) In the non-degenerate case :

$$P_D |n\rangle = |n^0\rangle.$$

If we substitute $|n\rangle = |n^0\rangle + \alpha |n^1\rangle + \alpha^2 |n^2\rangle$

and $\Delta_n = \alpha E_n^{(1)} + \alpha^2 E_n^{(2)}$ we get-

At $O(\alpha)$ in eq. of part (d)

$$\langle n^0 | V | n^0 \rangle = \alpha E_n^{(1)} \Rightarrow E_n^{(1)} = \langle n^0 | H_1 | n^0 \rangle.$$

At $O(\alpha^2)$ in eq. of part (d)

$$\langle n^0 | \hat{H}_1 \frac{\phi_D}{E - H_0} (\hat{H}_1 - E_n^{(1)}) | n^0 \rangle = E_n^{(2)}$$

$$\Rightarrow \langle n^0 | \hat{H}_1 \frac{\phi_D}{E - H_0} \hat{H}_1 | n^0 \rangle = E_n^{(2)}$$

which matches the result

From class :

$$E_n^{(2)} = \sum_{k \neq n} \frac{|\langle n^0 | \hat{H}_1 | k^0 \rangle|^2}{E_n^0 - E_k^0}$$

In the degenerate case we wait-

$$P_D H_1 P_D P_D |n\rangle = E_n^{(1)} P_D |n\rangle.$$

Thus $P_D |n\rangle$ is an eigenstate of $P_D H_1 P_D$ with eigenvalue $E_n^{(1)}$. Again this is what we found in class to lowest order!

17.3.4 (1) We want to calculate $\langle n\ell m | \frac{1}{r} | n\ell m \rangle$.

Consider $H = \frac{p^2}{2\mu} + \left(-\frac{e^2}{r} \right)$

The exact energy levels are given by.

$$E_n = - \frac{\mu(e^2)^2}{2\hbar^2} \frac{1}{n^2}$$

(see eq. (13.1.19) and (13.1.20)).

The first order correction upto λ is

$$E_n^I = \frac{\mu e^2 \lambda}{\hbar^2} \cdot \frac{1}{n^2}$$

$\therefore$ From our rules of perturbation theory we see that-

$$\langle n\ell m | \frac{1}{r} | n\ell m \rangle = \frac{\mu e^2}{\hbar^2 n^2}$$

(Hope you understand why the degeneracy of $|n\ell m\rangle$ states does not matter!).

(2) Now Consider

$$H = \frac{p^2}{2\mu} - \frac{e^2}{r} + \frac{\lambda}{r^2}$$

Since this modifies the centrifugal term the exact energy levels are

$$E_{k,\ell} = \frac{-\mu e^4}{2\hbar^2 (k+\ell'+1)^2}$$

(6)

where ℓ' is obtained from

$$\frac{\hbar^2 \ell'(\ell'+1)}{2\mu r^2} = \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} + \frac{\lambda}{r^2}.$$

$$\frac{\hbar^2}{2\mu r^2} \left[(\ell'+1) \frac{d\ell'}{d\lambda} + \ell' \frac{d\ell'}{d\lambda} \right] \bigg|_{\lambda=0} = \frac{1}{r^2}$$

$$\frac{\hbar^2}{2\mu} (2\ell+1) \left(\frac{d\ell'}{d\lambda} \right) \bigg|_{\lambda=0} = 1$$

$$\therefore \left(\frac{d\ell'}{d\lambda} \right) \bigg|_{\lambda=0} = \frac{2\mu}{\hbar^2} \frac{1}{(2\ell+1)}.$$

$$E_{\kappa, \ell}(\lambda) = -\frac{\mu e^4}{2\hbar^2 n^2} + \frac{\mu e^4 \lambda}{\hbar^2 n^3} \left(\frac{d\ell'}{d\lambda} \right) \bigg|_{\lambda=0} + O(\lambda^2)$$

First order correction

$$E'_{\kappa, \ell} = \lambda \frac{\mu e^4}{\hbar^2 n^3} \cdot \frac{2\mu}{\hbar^2} \frac{1}{(2\ell+1)}$$

If we set $\frac{\hbar^2}{\mu e^2} = a_0$, then we see that-

$$\left\langle \frac{1}{r^2} \right\rangle_{n\ell m} = \frac{1}{a_0^2} \frac{1}{n^3} \frac{1}{(\ell + \frac{1}{2})}.$$

(3) In order to evaluate $\left\langle \frac{1}{r^3} \right\rangle_{n\ell m}$ we note that-

$$\langle [\hat{H}, \hat{P}_r] \rangle_{n\ell m} = 0 \quad \text{for any operator } \hat{P}_r.$$

If $P_r = -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right)$ then

$$\hat{H} \equiv \left\{ \frac{p_r^2}{2\mu} + \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{e^2}{r} \right\}$$

$$[\hat{H}, \hat{p}_r] = \left[\left(\frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{e^2}{r} \right), \hat{p}_r \right]$$

$$= \left[\left(\frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{e^2}{r} \right), -i\hbar \frac{\partial}{\partial r} \right]$$

$$= -i\hbar \left[\left(\frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{e^2}{r} \right), \frac{\partial}{\partial r} \right]$$

$$= -i\hbar \left\{ -\frac{\hbar^2 \ell(\ell+1)}{\mu r^3} + \frac{e^2}{r^2} \right\}$$

$$\left\langle \left\{ -\frac{\hbar^2 \ell(\ell+1)}{\mu r^3} + \frac{e^2}{r^2} \right\} \right\rangle_{n\ell m} = 0$$

$$\Rightarrow \left\langle \frac{1}{r^3} \right\rangle_{n\ell m} = \underbrace{\frac{\mu e^2}{\hbar^2}}_{\downarrow \frac{1}{a_0}} \frac{1}{\ell(\ell+1)} \left\langle \frac{1}{r^2} \right\rangle_{n\ell m}$$

$$\Rightarrow \left\langle \frac{1}{r^3} \right\rangle_{n\ell m} = \frac{1}{a_0} \frac{1}{\ell(\ell+1)} \left\langle \frac{1}{r^2} \right\rangle_{n\ell m}$$

From part (2) we get-

$$\left\langle \frac{1}{r^3} \right\rangle_{n\ell m} = \frac{1}{a_0^3} \frac{1}{n^3} \frac{1}{\ell(\ell+1)} \left(\ell + \frac{1}{2} \right)$$

(4) To find the mean kinetic energy consider

(8)

$$H = \frac{P^2}{2\mu} - \frac{e^2}{r} + \frac{\lambda P^2}{2\mu}$$
$$= \frac{P^2}{2\mu} (1+\lambda) - \frac{e^2}{r}.$$

The exact energy levels are

$$E_n = - \frac{\mu e^4}{2\hbar^2(1+\lambda)} \frac{1}{n^2}.$$

First order correction then gives

$$\langle T \rangle \equiv \left\langle \frac{P^2}{2\mu} \right\rangle_{n\ell m} = \frac{\mu e^4}{2\hbar^2} \frac{1}{n^2}$$

From part (1) we see that -

$$\langle V \rangle = - \left\langle \frac{e^2}{r} \right\rangle = - \frac{\mu e^4}{\hbar^2 n^2}$$

Thus we see that

$$\langle T \rangle = - \frac{1}{2} \langle V \rangle$$

which is the virial theorem (see Exercise (7.4.3) and Exercise 13.1.5).